TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Crease-resistant Cotton Fabrics

Crosslinking cellulose with DMDHEU so cotton springs back from a crease — and measuring what that recovery costs in strength.

WP-4202 · TEXTILE FINISHING SESSIONAL · EXPERIMENT 3 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to produce a **crease-resistant (easy-care / durable-press) finish** on cotton by pad–dry–cure application of DMDHEU resin with a magnesium chloride catalyst; to calculate resin add-on and catalyst dose from first principles; to cure under controlled conditions; and to assess the result by wrinkle-recovery behaviour while recording the well-known cost of the finish — loss of tensile and, especially, tear strength.

Why cotton creases. A cotton fibre bent sharply holds the bend because, in the amorphous regions of cellulose, hydrogen bonds break under stress and re-form in the new, bent configuration — the fibre has no internal spring pushing it back. Wool and polyester recover because they possess resilient covalent networks (disulfide crosslinks; entangled thermoplastic chains with a set shape). Cotton has neither. Creasing is worse when wet, because water plasticises cellulose and makes hydrogen bonds still more mobile — which is why cotton crumples in the wash and why “dry recovery” and “wet recovery” are distinguished.

How resin finishing fixes it. A crosslinking resin introduces *covalent* bridges between neighbouring cellulose chains inside the fibre. When the fibre is bent, these crosslinks are stretched but cannot re-form elsewhere; on release they pull the chains back to the configuration in which the finish was cured — the flat state. The standard agent is **DMDHEU** (dimethylol dihydroxyethyleneurea), an N-methylol compound whose two –CH_2OH groups condense with cellulose –OH groups under acid catalysis and heat. DMDHEU dominates because it gives high recovery with better strength retention and lower chlorine retention than older urea–formaldehyde or melamine resins. Its known drawback is formaldehyde release (from unreacted methylol groups and hydrolysis), managed industrially by etherified/modified (low- and zero-formaldehyde) DMDHEU grades, formaldehyde scavengers, and optimised curing.

The catalyst and the cure. The condensation needs a latent acid catalyst — here **$MgCl_2 \cdot 6H_2O$**, often moderated with citric acid in industrial recipes. Too little catalyst or too cool a cure leaves resin

unreacted (poor recovery, high free formaldehyde, finish washes out). Too hot or too long a cure, or too much catalyst, attacks the cellulose itself: chains are cut by acid hydrolysis, and strength — tear strength first and worst — falls. The whole experiment is a controlled walk along that trade-off curve. Softener and, in industrial practice, a polyethylene emulsion are added to the bath largely to buy back tear strength and abrasion resistance at a given recovery level.

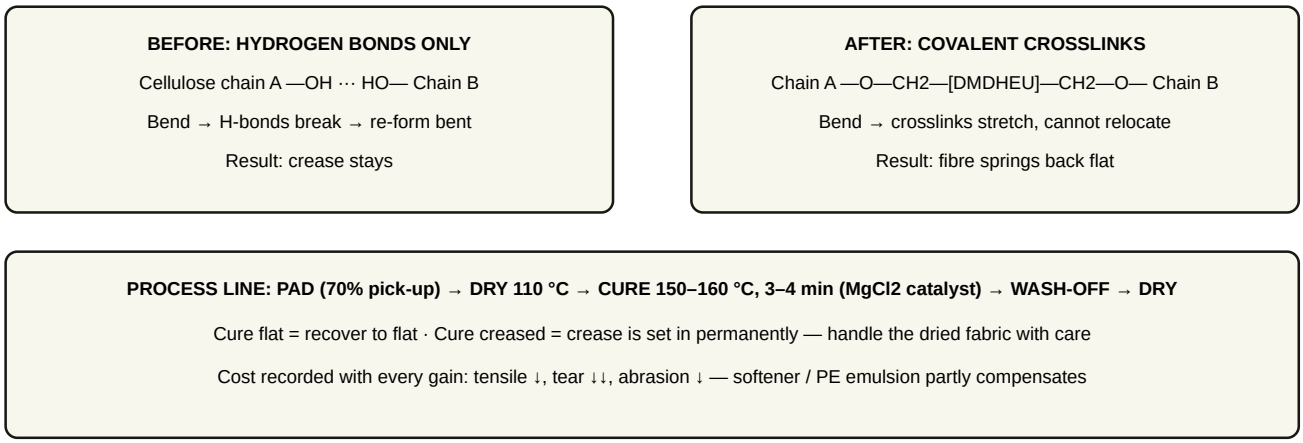


Fig. 1 — Top: cellulose–resin crosslinking schematic — hydrogen bonds relocate, covalent crosslinks cannot. Bottom: the pad–dry–cure sequence and its strength cost.

2. Apparatus, materials and safety

ITEM	SPECIFICATION / USE
Laboratory padder, stenter/oven	As Experiments 1–2; oven must hold 150–160 °C accurately — verify with a thermocouple or temperature strips
Fabric	Scoured and bleached cotton poplin, about 120 g/m², pieces 30 cm × 20 cm; warp direction marked on every piece
DMDHEU resin (45% solids type)	Crosslinking agent — dose in the recipe is as-supplied product
$MgCl_2 \cdot 6H_2O$	Catalyst; weighed as a percentage on resin weight
Additives	Non-ionic wetting agent 1 g/L; silicone or PE softener 20 g/L (tear/abrasion compensation)
Assessment kit	Wrinkle-recovery tester (or folded-specimen + weight + protractor rig), standard creasing load, stopwatch; tensile/tear results recorded comparatively if a tester is available

Safety. N-methylol resins release formaldehyde, especially during curing: gloves, ventilation at the oven, and no eating or drinking in the finishing laboratory. $MgCl_2$ bath is mildly acidic when hot — treat splashes as for dilute acid. Keep the cured fabric out of face contact until after the wash-off.

3. Recipe and procedure

Recipe

COMPONENT	AMOUNT	FUNCTION
DMDHEU resin (as supplied)	50 g/L	Crosslinker
$MgCl_2 \cdot 6H_2O$ catalyst	7.5 g/L	15% on resin weight — see calculation below
Softener (silicone/PE type)	20 g/L	Handle; partly restores tear and abrasion performance
Wetting agent	1 g/L	Even penetration
Wet pick-up target	70%	Gives 3.5% owf resin — see calculation below

Procedure

1. Mark the warp on each piece, condition and weigh dry. Prepare an **untreated control** piece that travels through drying only — every gain in Section 5 is measured against it.
2. Make up 300 mL of bath: water, wetting agent, resin, softener, and the catalyst solution **last**. Once catalyst is in, the bath has a working life (pot life) of a few hours — do not prepare it the day before.
3. Pad each piece twice, checking wet pick-up by weighing; adjust the nip to 70% ± 3%.
4. Dry at 100–110 °C for 2–3 min, keeping the fabric **flat and uncreased**. The state in which the fabric is cured is the state it will try to return to.
5. Cure at 150–160 °C for 3–4 min exactly (record the actual temperature and time for each batch). Start the timer only when the oven has recovered to temperature after loading.
6. Wash off: warm rinse with 2 g/L soda ash, then water — this removes unreacted resin, catalyst salts and surface formaldehyde. Dry flat, condition 24 h at standard atmosphere before testing.
7. Optional comparison (highly instructive if time allows): cure one extra padded piece deliberately folded. It returns from testing with the fold set in — direct proof that crosslinks memorise the cured configuration.

4. Worked calculations

Catalyst dose — 15% on resin weight

$$MgCl_2 \cdot 6H_2O = 15\% \times 50 \text{ g/L} = 7.5 \text{ g/L}$$

Catalyst is always dosed on the resin, never guessed as an independent number: scale the resin for a stronger or softer finish and the catalyst must scale with it, or the cure chemistry changes twice over.

Resin add-on

$$\text{Resin add-on} = 50 \text{ g/L} \times 0.70 = 35 \text{ g/kg} = 3.5\% \text{ owf}$$

For a 7.2 g poplin piece (30 cm × 20 cm at 120 g/m²) the resin carried is 35 × 0.0072 = 0.25 g of as-supplied product. Note the convention: textile recipes quote *product* add-on; the active DMDHEU solids are lower (about 45% of product weight for the common grade), and a rigorous record states both.

What a change of recipe would do (worked check)

Raising resin to 80 g/L at the same 70% pick-up gives 80 × 0.70 = 56 g/kg (5.6% owf) with catalyst 15% × 80 = 12 g/L. Recovery would rise — but so would strength loss and formaldehyde release. The standard 50 g/L recipe in this sheet is deliberately a *balanced* apparel recipe, not the maximum-recovery recipe.

5. Assessment and results

The bench assessment is the **wrinkle-recovery angle**: a folded specimen is creased under a standard load for a set time, released, and the angle to which it opens after recovery is measured in the warp and weft directions. The reported figure is the sum, W + F. Full standard method (AATCC 66 principle) and the appearance-based DP rating (AATCC 124) are treated in Experiment 4; here the same rig is used comparatively, treated against control, with identical folding, load and recovery time for every specimen — the comparison is only valid if nothing but the fabric changes.

SPECIMEN	WRINKLE-RECOVERY ANGLE, W + F (TYPICAL RECORDED)	OBSERVATION
Untreated cotton control	≈ 156°	Crease sharp and persistent; poor recovery wet and dry
DMDHEU-treated (this recipe)	≈ 288°	Specimen opens almost flat; crease faint after recovery

Gain from finishing	288° − 156° = 132°	Recovery improved by 132° (sum of warp + weft)

$$\text{Gain} = 288° - 156° = 132°$$

The strength cost — record it, do not hide it. Crosslinking embrittles cellulose: chains that can no longer slip and share load fail one by one under tearing stress. At this recipe level, expect typically a modest tensile loss (of the order of 10–20%) and a distinctly larger tear-strength loss (often 25–40% without compensation) plus reduced abrasion resistance; the softener/PE component in the bath exists to claw part of that back, mainly on tear and abrasion. If tensile/tear testers are available, break treated and control strips in both directions and enter the real percentages in the record; if not, state clearly that strength was assessed qualitatively in this session and quantitative testing follows the Experiment 4 pattern. A finish report that quotes the 132° gain without the strength column is incomplete work.

OPTIMISATION, IN ONE PARAGRAPH

Recovery angle, strength retention, formaldehyde release and handle pull in different directions. More resin / hotter cure: recovery up, strength and formaldehyde worse. Less resin / cooler cure: strength kept, recovery and durability down. The finisher's skill is choosing the point on that trade-off that the end use (shirting vs workwear vs sheeting) actually needs — and proving it with both columns of numbers.

6. Precautions

- Add catalyst last, use the bath promptly, and discard leftovers — a catalysed resin bath left standing pre-condenses and gives streaky, weak results the next day.
- Keep the dried fabric perfectly flat before and during cure; any accidental fold is crosslinked in as a permanent crease.
- Time the cure at temperature, not from oven-door closing; under-cure is the commonest cause of “the finish washed out”.
- Wash off after cure before assessment or handling; surface resin otherwise inflates recovery readings and free-formaldehyde complaints.
- Test treated and control specimens in the same session, same load, same recovery time, warp marked — recovery angles are comparative numbers.

7. Practice (viva preparation)

1. Explain creasing of cotton at the molecular level and state exactly what a crosslink changes.

2. The resin dose is raised to 60 g/L at 70% pick-up. Calculate the new add-on (% owf) and the new catalyst dose at 15% on resin weight.
3. Why is tear strength hurt more than tensile strength by crosslinking?
4. A batch shows good recovery on the day but poor recovery after five washes. Give the two most likely process causes.
5. Why must the fabric be flat during curing? Answer in terms of crosslink memory.
6. Name the main health/regulatory concern of DMDHEU finishing and two ways industry reduces it.

Answer key

1. In bent cellulose, hydrogen bonds in amorphous regions break and re-form in the bent shape, so there is no restoring force. A DMDHEU crosslink is a covalent bridge fixed in the cured (flat) configuration; bending stretches it elastically and it pulls the chains back on release.
2. Add-on = 60 × 0.70 = 42 g/kg = 4.2% owf. Catalyst = 15% × 60 = 9.0 g/L $MgCl_2 \cdot 6H_2O$.
3. In tearing, load concentrates on a few yarns/fibres at the tear tip and survival depends on chains slipping to share stress. Crosslinks prevent that slip, so fibres at the tip break successively at low load. In a tensile test the load is already shared across the specimen, so the loss is smaller.
4. Under-curing (temperature or time short, or catalyst under-dosed), so crosslinks never fully formed; or omission/ineffectiveness of the wash-off is not the cause of loss — the second likely cause is hydrolysis of crosslinks from an over-acid bath / excessive catalyst attacking cellulose, giving apparent recovery that collapses as damaged resin–cellulose bonds break in washing. Either way: check cure records and bath pH first.
5. Crosslinks form in whatever geometry the fabric holds during cure and thereafter restore that geometry. Flat at cure = recovery to flat; folded at cure = a permanently set crease.
6. Formaldehyde release (skin sensitisation; regulated limits on skin-contact textiles). Reduced by etherified/modified low-formaldehyde DMDHEU, formaldehyde scavengers, optimised (complete) curing, and thorough after-washing.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib